CS 739: DISTRIBUTED SYSTEMS

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WHO AM I?

Professor Andrea Arpaci-Dusseau
PhD from U.C. Berkeley – Implicit Coscheduling of Parallel Jobs
Postdoc at Stanford – Taught OS course there

Teach CS 537, CS 736 (Adv OS), CS 739 (Dist Systems) and CS 402

Co-Advised about 30 PhD students, mostly research in file and storage systems

Co-Chair for 2018 OSDI Program Committee

2018 (w/ Remzi) - SIGOPS Mark Weiser award for "outstanding leadership, innovation, and impact in storage and computer systems research"

CALL ME

TODAY'S AGENDA

What are distributed systems?
What are the benefits and challenges?

What will you do in this course?

WHAT IS A DISTRIBUTED SYSTEM?

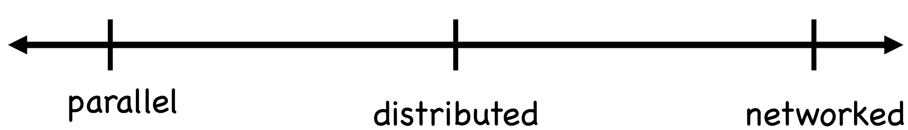
Leslie Lamport says:

More technical definition:

More detailed: (Google Code University – Distributed System Design)

How are parallel, distributed, networked systems different?

- All contain nodes (processing, memory, disk) connected with network



WHY BUILD DISTRIBUTED SYSTEMS?

WHY STUDY DISTRIBUTED SYSTEMS?

WHY IS IT CHALLENGING?

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HOW WILL YOU LEARN IN 739?

Two main components: Readings and Projects

Read Research Papers: Learn concepts + How to evaluate others' research

- Learn from the best of the past
- What has been tried, what has worked, what hasn't
- Research must be done at the right time given technology

Projects: How to do research yourself

- Sophocles: "One must learn the thing by doing, for though you think you know it, you have no certainty until you try"
- Harder to synthesize than criticize
- Complete a few initial assignments to learn system tools and skills

WHAT TOPICS WILL WE COVER?

Failures

- Why do systems fail?
- How do modern replicated services handle failures?

Communication: what is the right abstraction/primitive?

- RPC

Client-server file systems

- Example systems: NFS, AFS, Coda, HA-NFS, LBFS
- Techniques: Caching, Leases, Checksums

WHAT TOPICS WILL WE COVER?

Ordering

- how to reason about time and distributed state
- logical clocks, vector clocks, snapshots

Fault-tolerance: how to tolerate failures

- idea: replication
- primary-backup, chain replication, state machine replication

Consensus: Paxos, (VR), Raft

- more efficient: SpecPaxos
- practical systems: Chubby, ZooKeeper

Byzantine: Byzantine generals, PBFT

WHAT WILL WE COVER?

Consistency

- View of data at instant in time depending on previous operations
- Weak, causal, strong, existential, nil-externality

Case Studies: Distributed Storage

- P2P routing: Chord
- Google File System
- Dynamo (quorum-based)
- Bigtable

Other topics as time permits

PAPER READINGS

Purpose: Learn Content AND Learn how to evaluate papers on your own
- Reading is fundamental for every researcher
- Key to idea formation
- "Everything has been thought of before; the problem is to think of it again" – Goethe

1-2 (or 3) papers per lecture
- Papers are about 16 pages, dense
- Selected from "best of best" (SIGOPS Hall of Fame)
  - SOSP, OSDI
- Bring papers to lecture; be prepared to answer questions!

How to read a paper?
- Read many times to understand well
- Once to get basic ideas and terminology
- Second time to understand details
- Discuss questions with reading group
- Reference paper during lecture

WHAT SHOULD YOU LEARN FROM A PAPER?

What problem are the authors trying to solve?

- Why was the problem important?
- Why was the problem not solved by earlier work?

What is the authors' solution?

- How does their approach solve the problem?
- How is the solution unique and innovative?
- What are the details?

How do the authors evaluate their solution?

- What specific questions do they answer?
- What simplifying assumptions do they make?
- What is their methodology?
- What are their strengths and weaknesses?
- What is left unknown?

What do you think?

- Is the problem still important?
- Did the authors solve the stated problem?
- Did the authors adequately demonstrate that they solved the problem?

What future work does this research point to?

PAPER WRITE-UP BEFORE EACH LECTURE

Purpose

- Ensure you've read the paper before lecture

One question per paper or per lecture in Canvas

- May have peer reviews later

Grades: Simple 0-4 points

- Receiving a 3 is fine…

Reading Groups

Purpose

- Better understanding of paper content
- Be comfortable sharing your thoughts and questions
- Get to know more people in class

Beginning of semester: New random group every 2 weeks

- 6 people, meet with at least 1 other person
- Meet weekly or so to discuss readings
- Ideally before lecture, but fine if after too

Later in semester: Pick your own reading group

ASSESSMENTS: PART 1

Lecture Material (60%)

- Paper Write-Ups: 15%
- Reading Groups: 5%
- Class Participation (or other paper questions):10%
- Exam 1: 15%
- Exam 2: 15%

Closed book, multiple choice

Assess concepts discussed in class

ASSESSMENTS: PART 2

Projects (40%) – Prof. Remzi Arpaci-Dusseau

Four projects (3 well-defined, 1 more open-ended) with groups

Gain hands-on experience

1. Reliable Communication layer (Measurement)
   - Due in 2+ weeks
   - Random group of 4 students
   - Grade with 10-minute demo (code + graphs) to Prof. Remzi
   - Help on piazza and TAs
2. Client-Server File System (FUSE)
3. Replicated File System
4. Open-Ended Project

SUMMARY: LEARNING OUTCOMES FOR 739

- Understand fundamental challenges in distributed systems, such as availability, reliability, consistency, scalability
- Understand fundamental techniques used in distributed systems, such as those for ordering and consensus
- Experience with case studies of current distributed systems
- Deploy, measure, and modify existing distributed systems
- Implement open-ended research project and communicate results

NEXT LECTURE: FAILURE

Two papers

- Why do computers stop and what can be done about it? Jim Gray, 1985, SIGOPS Hall of Fame
- *Redundancy Does Not Imply Fault Tolerance: Analysis of Distributed Storage Reactions to Single Errors and Corruptions*, FAST 2017

Answer questions in Canvas

1. How does the **fault model** used in *Redundancy Does Not Imply Fault-Tolerance* compare to the failures described in *Why Do Computers Stop?*
2. How does the **expected behavior** tested for in *Redundancy Does Not Imply Fault-Tolerance* match the system behavior advocated in *Why do Computers Stop?*
3. For fault-tolerant storage, *Why do Computers Stop?* mentions that the **scope** of a failure can be limited. What did *Redundancy Does Not Imply Fault-Tolerance* discover about the scope of failures?

One paragraph for each question; 3-5 sentences per paragraph; full sentences!

WHAT TYPES OF FAILURES?

Halting failures:

Fail-stop:

Omission failures:

Network failures:

Network partition failure:

Timing failures:

Byzantine failures:

1. A temporal property of the system is violated (non-synchronized clocks, messages delayed)
2. Component stops with some notification to other components
3. Messages or operations lost with no notification (e.g., no buffer space)
4. Component simply stops with no notification except by timeout (e.g., no heartbeat msg)
5. Data corruption or loss, failures caused by malicious programs
6. A network fragments into two or more disjoint sub-networks
7. A network link breaks

WHAT TYPES OF FAILURES?

Halting failures:

Component simply stops with no notification except by timeout (e.g., no heartbeat msg)

Fail-stop:

Halting failure with some kind of notification to other components.

Omission failures:

Messages or operations lost with no notification (e.g., no buffer space)

Network failures:

A network link breaks

Network partition failure:

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Byzantine failures:

Data corruption or loss, failures caused by malicious programs

NEXT STEPS

Check out Canvas pages and Register on Piazza

Read two failure papers and answer questions in Canvas before lecture

Identify Members of Random Reading Group in Canvas

    Meet with portion of reading group by next Tuesday: See groups in Canvas

Get ready for First programming assignment: Due 2+ weeks

    Measurement and Implementation of Reliable Communication Layer